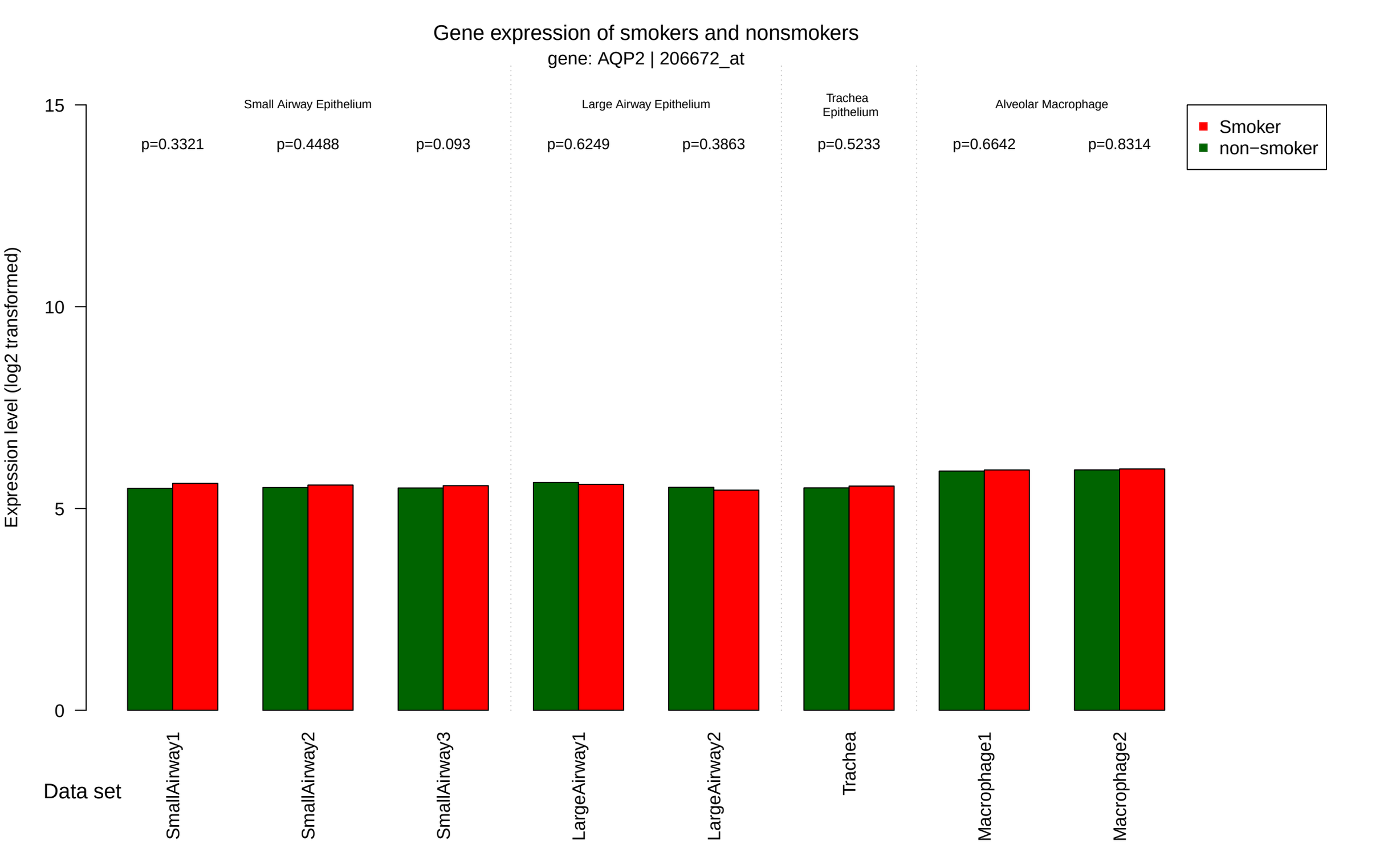
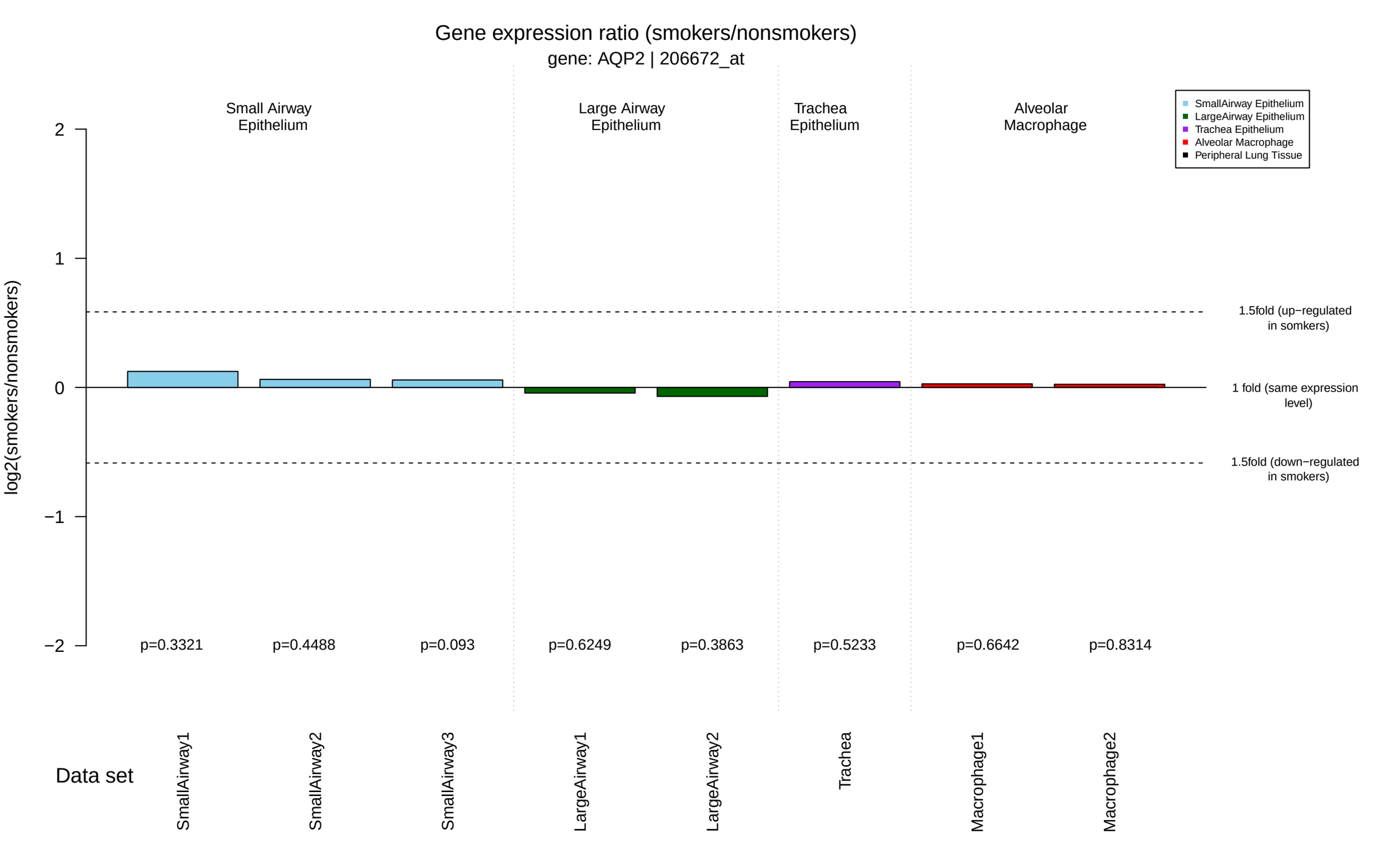
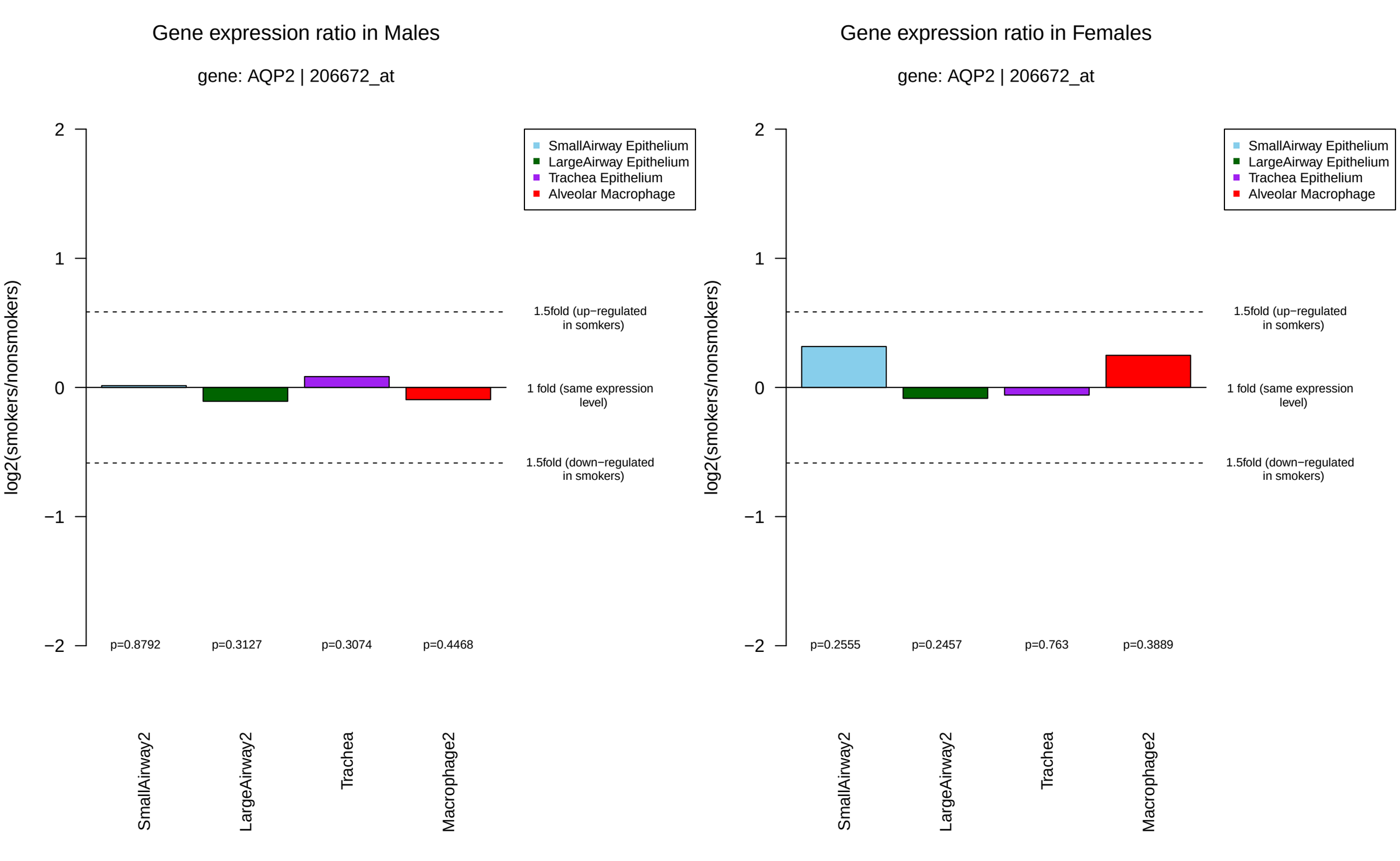
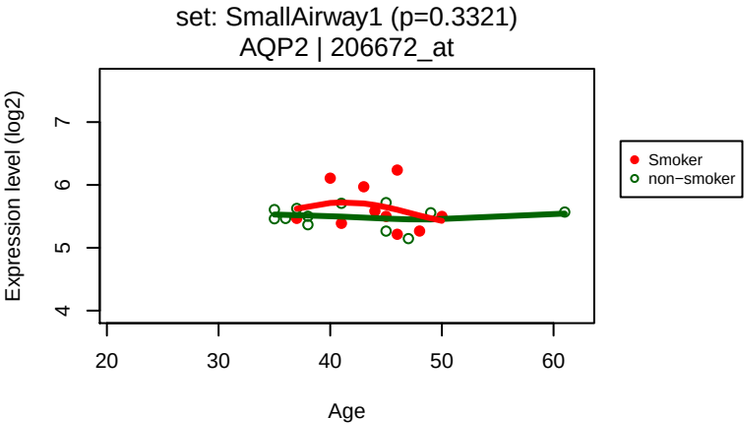
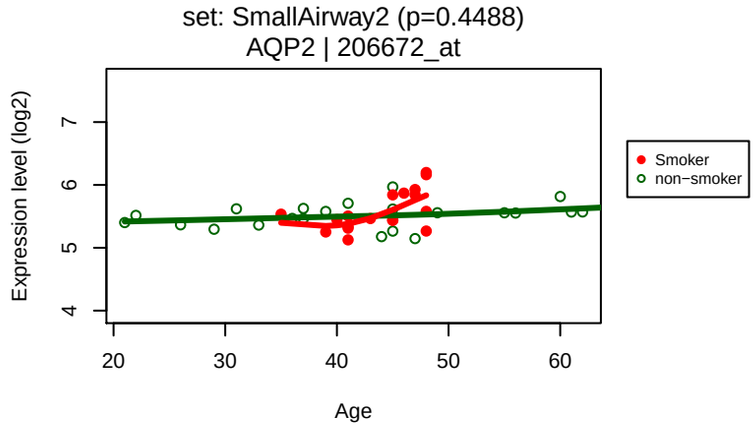
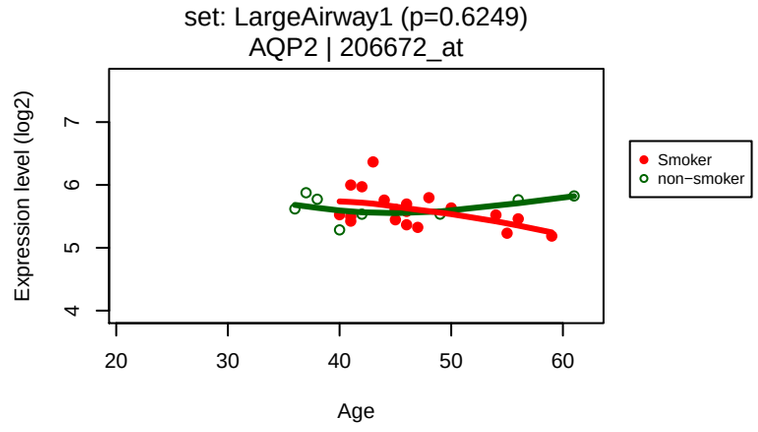

Gene expression of smokers and nonsmokers
gene: AQP2 | 206672_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3321
p=0.4488
p=0.093
p=0.6249
p=0.3863
p=0.5233
p=0.6642
p=0.8314
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

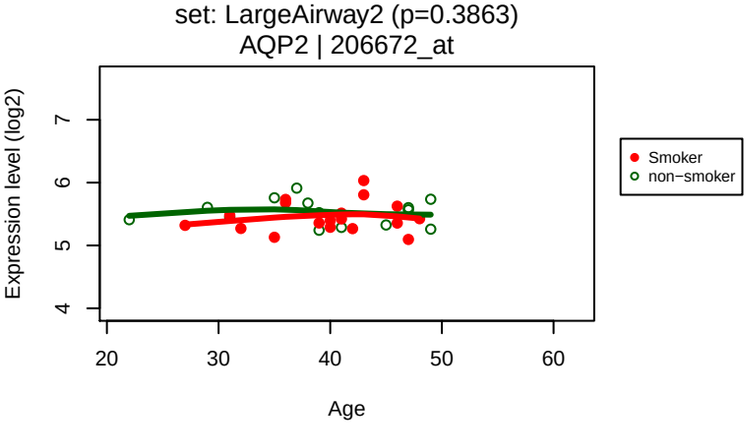

Gene expression ratio (smokers/nonsmokers)
gene: AQP2 | 206672_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3321
p=0.4488
p=0.093
p=0.6249
p=0.3863
p=0.5233
p=0.6642
p=0.8314
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

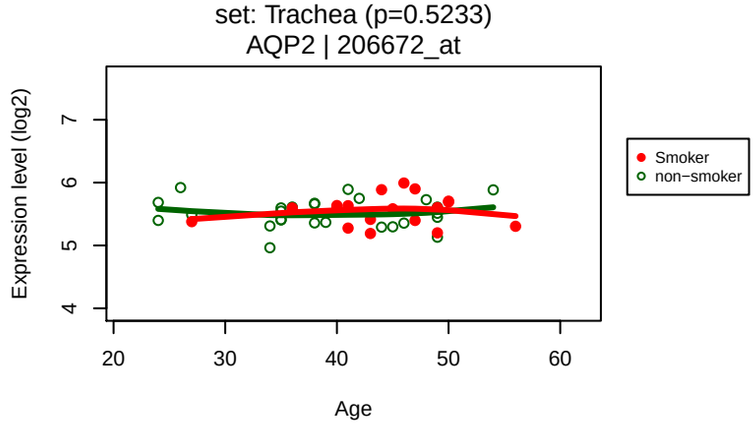

Gene expression ratio in Males
gene: AQP2 | 206672_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8792
p=0.3127
p=0.3074
p=0.4468
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP2 | 206672_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2555
p=0.2457
p=0.763
p=0.3889
SmallAirway2
LargeAirway2
Trachea
Macrophage2

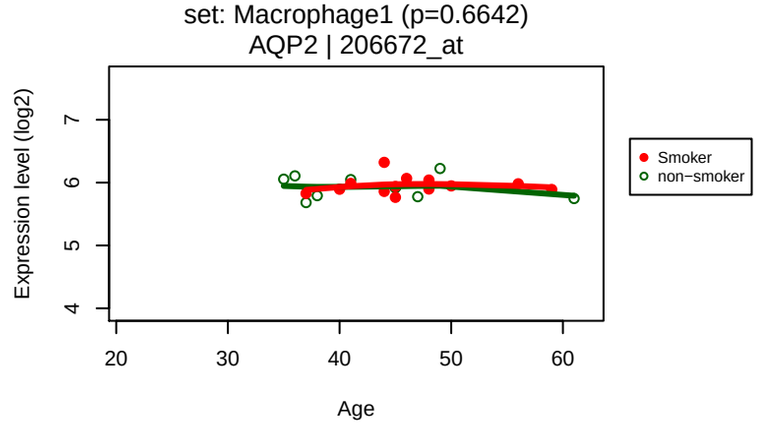

set: SmallAirway1 (p=0.3321)
AQP2 | 206672_at
Expression level (log2)
Age
Smoker
non−smoker

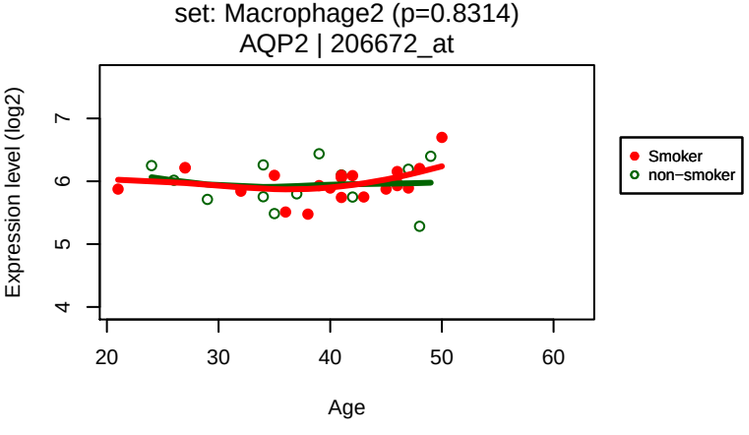

set: SmallAirway2 (p=0.4488)
AQP2 | 206672_at
Expression level (log2)
Age
Smoker
non−smoker

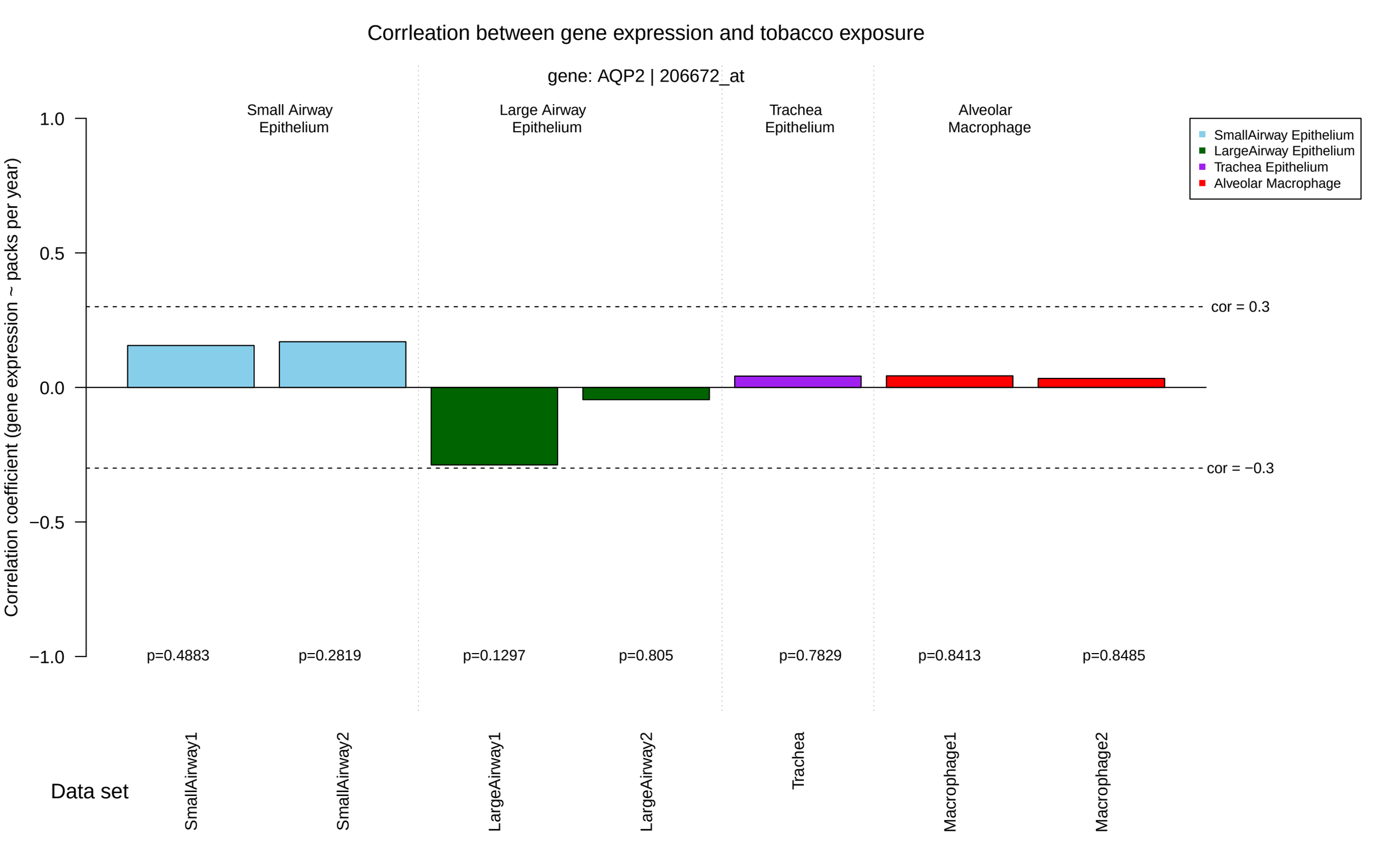

set: LargeAirway1 (p=0.6249)
AQP2 | 206672_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3863)
AQP2 | 206672_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5233)
AQP2 | 206672_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6642)
AQP2 | 206672_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8314)
AQP2 | 206672_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP2 | 206672_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4883
p=0.2819
p=0.1297
p=0.805
p=0.7829
p=0.8413
p=0.8485
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2